NEWTON SCHOOL OF TECHNOLOGY
Data Visualization & Analytics | Capstone Project

NYC Real Estate Market Analysis

A Data-Driven Study of NYC Rolling Sales 2016–2017

Project Title:
NYC_Real_Estate_Analysis

Team ID:
DVA-B-T16 | Section B

Sector:
Commercial Real Estate

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Faculty Mentors:
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Dataset Period:
2016 – 2017

Team Members

Role	Name	GitHub Username
Project Lead	Khushi / Krishnadevan M.L.	khushi-madan18 / ka-ori
Data Lead	Krishnadevan M.L.	Ka-ori
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1. Executive Summary

This report presents a comprehensive analysis of New York City's real estate market using the NYC Rolling Sales dataset covering the period from September 2016 to August 2017. The dataset, sourced from Kaggle, contains 33,167 processed property transactions across five boroughs: Manhattan, Brooklyn, Queens, the Bronx, and Staten Island.

The primary objective of this project is to identify which NYC boroughs and property types generate the highest sales performance, and how pricing trends and transaction patterns evolve over time. Through Python-based ETL pipelines and interactive Tableau dashboards, the team uncovered critical market dynamics that can inform investment decisions, pricing strategy, and policy planning.

25,491	$19.05B	$747.18K	43,335	$403.35
Total Transactions	Total Sales Volume	Average Sale Price	Total Units Sold	Avg Price / Sqft

Key findings reveal that Manhattan commands the highest average sale prices (exceeding $3.5M), while Brooklyn and Queens drive the greatest transaction volumes. Family homes dominate the market, accounting for over 80% of all transactions. Sales velocity averaged 2,124 transactions per month, with Winter recording the lowest seasonal activity. Buildings aged 75–105 years represent the most transacted segment of the market, pointing to the dominance of legacy construction in NYC's housing stock.

2. Business Problem & Objectives

2.1 Problem Context

New York City's real estate market is one of the largest and most complex in the world. With thousands of property transactions occurring every month across five boroughs and dozens of property categories, stakeholders face significant challenges in identifying actionable intelligence from raw data.

Real estate investors, brokers, and market analysts frequently encounter the following challenges:

- Difficulty identifying high-value boroughs and neighborhoods for targeted investment
- Lack of visibility into seasonal sales fluctuations and demand cycles
- Inability to compare property type performance across geographic segments
- Poor understanding of how building age, size, and tax classification affect pricing

2.2 Core Business Question

> ***Which NYC boroughs and property types generate the highest sales performance, and how do pricing trends and transaction patterns change over time?***

2.3 Project Objectives

- Analyze borough-wise average sale prices and price per square foot to identify premium markets
- Segment property types to quantify residential vs. commercial demand
- Map temporal trends in transaction volume and pricing across quarters and seasons
- Build interactive Tableau dashboards that enable decision-makers to explore the data dynamically
- Derive 3–5 actionable business recommendations supported by statistical evidence

3. Dataset Description

3.1 Data Source

The analysis is based on the NYC Rolling Sales Dataset publicly available on Kaggle. This dataset captures rolling 12-month property sale records across all five New York City boroughs, maintained by the NYC Department of Finance.

Attribute	Details
Source Name	NYC Rolling Sales Dataset / Kaggle
Raw Row Count	84,548 records
Processed Row Count	33,167 records
Raw Column Count	22 columns
Processed Column Count	19 columns
Time Period Covered	September 2016 – August 2017
File Format	CSV
Dataset Size	Approximately 4.2 MB (processed)

3.2 Key Columns

Column Name	Data Type	Description	Role in Analysis
borough	Integer (1–5)	NYC borough identifier	Borough-wise comparison
neighborhood	String	Area / locality name	Regional sub-analysis
building_class_category	String	Type of property / building	Property segmentation
sale_price	Float	Transaction value in USD	Primary KPI metric
sale_date	Date	Date of the transaction	Time-series analysis
gross_square_feet	Float	Total floor area	Price-per-sqft calculation
year_built	Integer	Year of construction	Building age analysis
residential_units	Integer	Number of residential units	Residential demand metrics
commercial_units	Integer	Number of commercial units	Commercial demand metrics
tax_class_at_present	String	Current property tax class	Tax segment analysis

4. ETL Pipeline & Data Cleaning

4.1 Pipeline Overview

The project follows a structured Extract, Transform, and Load (ETL) pipeline implemented across five Jupyter notebooks and an optional Python script. The pipeline was designed to be reproducible, documented, and version-controlled via GitHub.

Stage	Notebook	Description
01 – Extraction	01_extraction.ipynb	Load raw CSV; initial schema inspection; data profiling
02 – Cleaning	02_cleaning.ipynb	Handle nulls, outliers, type conversion, deduplication
03 – EDA	03_eda.ipynb	Exploratory analysis; distribution plots; correlation heatmaps
04 – Statistical Analysis	04_statistical_analysis.ipynb	KPI computation; hypothesis testing; segment analysis
05 – Load Prep	05_final_load_prep.ipynb	Final feature engineering; export processed CSV for Tableau

4.2 Cleaning Steps

The raw dataset of 84,548 records was reduced to 33,167 high-quality records through the following transformations:

Cleaning Action	Rationale	Records Affected
Removed zero-value sale prices	Non-market transfers (gifts, inter-family)	~48,000 rows removed
Dropped null sale dates	Essential for time-series accuracy	<100 rows
Standardized borough encoding	Converted integer codes to names	All 33,167 rows
Removed extreme outliers	Transactions >99th percentile distort averages	~600 rows
Parsed and typed sale_date	Required datetime format for Tableau	All rows
Imputed missing square footage	Median imputation by building class	~2,400 rows

Cleaning Action	Rationale	Records Affected
Feature engineering	Created sale_year, sale_quarter, sale_season, building_age, price_per_sqft	All rows

4.3 Feature Engineering

Several derived columns were added to enable richer Tableau analysis:

- sale_year – extracted from sale_date for year-over-year comparison
- sale_quarter – Q1/Q2/Q3/Q4 labelling for quarterly trend charts
- sale_season – Spring / Summer / Fall / Winter based on sale month
- building_age – calculated as sale_year minus year_built
- price_per_sqft – sale_price divided by gross_square_feet
- price_category – binned into Low / Mid-Low / Mid-High / High tiers
- property_type_group – mapped building_class_category to five groups: Family Home, Commercial, Rental Apartment, Vacant Land, Other

5. KPI Framework

The KPI framework was designed to translate raw transaction data into decision-relevant business metrics. Each KPI is computed at both the aggregate level and at the borough/property-type level to enable cross-segment comparison.

KPI	Definition	Formula	Computed Value
Total Transactions	Total distinct property sales	COUNT(*)	25,491
Total Sales Volume	Aggregate transaction value	SUM(sale_price)	$19.05 Billion
Average Sale Price	Mean transaction value	AVG(sale_price)	$747,180
Avg Price Per Sqft	Price efficiency per area unit	SUM(price) / SUM(sqft)	$403.35
Total Units Sold	All residential + commercial units	SUM(total_units)	43,335
Sales Velocity	Monthly transaction rate	COUNT / months	2,124 / month
Residential Unit Total	Total residential unit count	SUM(residential_units)	41,830
Commercial Unit Total	Total commercial unit count	SUM(commercial_units)	1,533
Average Building Age	Mean age of transacted buildings	AVG(building_age)	75.48 years

6. Exploratory Data Analysis

6.1 Borough-Level Overview

Borough analysis reveals stark pricing disparities across Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Manhattan commands a price per square foot exceeding $1,200, more than three times that of any other borough. Brooklyn and Queens, while more affordable, drive the highest transaction volumes in the city.

Borough	Avg Sale Price	Avg $/Sqft	Transactions	Share of Volume
Manhattan	$3.5M+	$1,200+	~3,200	~38%
Brooklyn	$1.1M	$420	~7,100	~29%
Queens	$620K	$310	~8,900	~22%
Bronx	$420K	$220	~4,200	~7%
Staten Island	$480K	$240	~2,100	~4%

6.2 Property Type Distribution

Family homes dominate the NYC sales landscape, accounting for over 80% of all transaction counts. Rental apartments and commercial properties combined constitute less than 10% of transactions by count, though commercial units command significantly higher per-unit values.

Property Type Group	Transaction Count	% of Total	Key Insight
Family Home	~22,500	88.3%	Dominant category across all boroughs
Rental Apartment	~1,200	4.7%	Concentrated in Manhattan and Brooklyn
Commercial	~500	2.0%	Small count but high average value
Other	~700	2.7%	Mixed-use and specialty properties
Vacant Land	~350	1.4%	Mostly in outer boroughs

6.3 Sale Price Distribution

The sale price distribution is right-skewed, with the majority of transactions occurring in the $300K–$600K range. The histogram from Dashboard 1 shows the modal price bin at $400K–$500K, which aligns with the mid-range pricing segment containing the highest transaction

density. Ultra-premium transactions above $1M represent a smaller but disproportionately valuable share of total sales volume.

Price Bin	Transaction Count	Category
$100K – $200K	~1,600	Low
$200K – $300K	~2,700	Low–Mid
$300K – $400K	~2,700	Mid-Low
$400K – $500K	~3,950	Mid (Peak)
$500K – $600K	~3,400	Mid-High
$600K – $700K	~2,900	High
$700K – $800K	~2,200	High
$800K – $900K	~1,800	Premium

6.4 Seasonal Sales Analysis

Sales are distributed relatively evenly across the four seasons, with Fall and Spring showing slightly elevated activity. Winter records the lowest transaction count and is identified as the weakest sales season. This pattern is consistent with broader US real estate market behavior, where cold-weather months dampen buyer activity.

Season	Transaction Share	Sales Volume	Activity Level
Spring	~26%	High	Strong buyer demand
Fall	~27%	High	Highest seasonal share
Summer	~25%	Moderate	Steady activity
Winter	~22%	Lowest	Off-peak period

6.5 Building Age Analysis

The building age distribution reveals that NYC's sales market is dominated by properties constructed 75–105 years ago, corresponding to the construction boom of the 1910s–1940s. Properties in the 90–105 year age range record the highest transaction count (over 6,000), highlighting the city's historic housing stock as the backbone of its real estate market.

7. Statistical Analysis

7.1 Price Trend Analysis

The average sale price trend line (Dashboard 1) remained broadly stable throughout the observation period, oscillating between $700K and $850K. A modest upward drift is visible from Q1 2017 onward, suggesting gradual market appreciation. The trough observed in early 2017 likely reflects typical post-holiday market slowdown.

7.2 Quarterly Transaction Patterns

Quarterly transaction counts from Dashboard 3 reveal that 2017 Q1 recorded the highest transaction volume at approximately 6,100 deals. Q3 of 2016 shows low count as it represents only a partial quarter in the dataset. The consistent Q2 and Q3 peaks confirm that mid-year months drive the majority of NYC real estate activity.

Quarter	Transaction Count	vs. Prior Quarter	Market Signal
2016 Q3 (partial)	~2,400	N/A – partial data	Dataset entry point
2016 Q4	~6,600	+175% (full quarter)	Year-end surge
2017 Q1	~6,100	-7%	Strong start
2017 Q2	~6,500	+7%	Peak activity
2017 Q3 (partial)	~3,800	–	Dataset cutoff

7.3 Tax Class Analysis

Tax Class 1 (primarily 1–3 family residential properties) dominates transaction counts at over 20,000 records, confirming the residential nature of most NYC transactions. Tax Class 2 (multi-unit apartment buildings), 2A, 2B, and Class 4 (commercial) collectively account for the remaining minority of sales.

7.4 Correlation Insights

Key statistical correlations identified during the analysis:

- Strong positive correlation between gross_square_feet and sale_price ($r \approx 0.68$) across all boroughs
- Moderate negative correlation between building_age and sale_price in Manhattan (older prestige buildings are an exception)

- Residential unit count positively correlated with total sales volume in Brooklyn and Queens
- Commercial unit presence shows weak correlation with overall borough volume due to low sample size

8. Tableau Dashboard Analysis

Three interactive Tableau dashboards were published on Tableau Public. Each dashboard targets a distinct analytical perspective and supports a specific set of user decisions. The dashboards are linked and share common filters including Borough Name, Property Type Group, Sale Season, Price Category, and Sale Quarter.

Dashboard URL: https://public.tableau.com/views/DVA_Capstone_17773722247150/Dashboard1

8.1 Dashboard 1 – Market Overview

Dashboard 1 provides a high-level market summary with five headline KPIs: Total Transactions (25,491), Total Sales Volume ($19.05B), Average Sale Price ($747.18K), Total Units Sold (43,335), and Average Price Per Sqft ($403.35). Supporting visualizations include a borough-wise average price per sqft bar chart, a sales count donut chart, an average sale price trend line across months, an average sale price by borough bar chart, and a sale price distribution histogram.

- Key finding: Manhattan's avg $/sqft exceeds all other boroughs combined
- Key finding: Brooklyn and Queens together account for over 60% of transaction counts
- Key finding: The $400K–$500K bin holds the highest transaction density

8.2 Dashboard 2 – Property Type Analysis

Dashboard 2 deep-dives into property type and building class segmentation. It features sales count by tax class, total units by borough and property type stacked bar chart, a season-wise sales volume donut, a transactions by property type horizontal bar, and a property type distribution treemap by borough. The average building age KPI of 75.48 years and the most common building class (01 One Family Dwellings) are prominently featured.

- Key finding: Tax Class 1 dominates all other classes by a factor of 10x
- Key finding: Queens has the highest total unit count, driven by large residential blocks
- Key finding: Family homes represent over $6.6B in sales volume in both Brooklyn and Queens each

8.3 Dashboard 3 – Time Trend & Price Analysis

Dashboard 3 examines temporal patterns and price segmentation. A monthly sales heatmap by year visualizes activity levels across months for 2016 and 2017. A quarterly transaction count bar chart reveals the 2017 Q1 peak. Sales count by price category shows Mid-Low pricing leads all categories. The building age distribution histogram confirms 90–105 year buildings as the dominant cohort. Sales Velocity of 2,124 per month is displayed as a headline KPI.

- Key finding: 2017 recorded higher overall activity than 2016 across all months
- Key finding: Winter is consistently the lowest-activity season
- Key finding: Mid-Low price category has the highest distinct sale price count

9. Key Insights

The following 10 insights are stated in decision language and are directly supported by the Tableau dashboards and statistical analysis:

#	Insight	Supporting Evidence	Implication
1	Manhattan is the premium pricing hub with avg sale price exceeding $3.5M	Dashboard 1 – Avg Sale Price by Borough	Target luxury investment in Manhattan for highest ROI
2	Brooklyn and Queens drive transaction volume, not pricing premiums	Dashboard 1 – Sales Count Donut	Volume-focused strategies should target outer boroughs
3	Family homes account for 88%+ of all transactions citywide	Dashboard 2 – Transactions by Property Type	Residential marketing investment yields the broadest market reach
4	Mid-Low price category ($300K–$500K) records the highest transaction density	Dashboard 3 – Sales Count by Price Category	Affordable inventory generates the most market activity
5	Winter is the consistently weakest season for real estate sales	Dashboard 3 – Sales Velocity KPI; Dashboard 2 – Season Donut	Reduce listing activity in Winter; concentrate campaigns in Fall/Spring
6	2017 sales volume exceeded 2016 across nearly all months	Dashboard 3 – Monthly Heatmap	Market was in an upward cycle during the study period
7	2017 Q1 recorded peak quarterly transactions (~6,100)	Dashboard 3 – Quarterly Transaction Count	January–March is a strategically important window for deal closings
8	Buildings aged 90–105 years dominate the transaction landscape	Dashboard 3 – Building Age Distribution	Renovation and redevelopment of legacy stock is a high-opportunity segment
9	Manhattan's avg $/sqft exceeds $1,200, more than 3x the next borough	Dashboard 1 – Avg $/Sqft by Borough	Price-per-sqft is the most discriminating metric for borough comparison
10	Commercial properties are few in count but represent high-value outliers	Dashboard 2 – Property Type Distribution Treemap	Selective commercial acquisition in Manhattan can yield outsized returns

10. Business Recommendations

Based on the 10 insights above, the following five actionable recommendations are proposed for real estate investors, brokers, and market analysts operating in the NYC market:

#	Linked Insight	Recommendation	Expected Impact
1	Manhattan premium pricing (Insight #1, #9)	Prioritize luxury investment in Manhattan's premium segments. Focus acquisition targeting on high-sqft properties in Chelsea, Midtown, and the Upper East Side where price appreciation trends are strongest.	Higher ROI potential; $3.5M+ avg sale values provide strong capital growth trajectory
2	Family homes dominate (Insight #3)	Increase residential property marketing spend, particularly for 1–3 family dwellings in Brooklyn and Queens. Develop buyer outreach programs for mid-range residential buyers.	Captures 88% of market demand; aligns marketing resources with highest-volume segment
3	Older buildings dominate sales (Insight #8)	Develop a renovation and redevelopment investment strategy targeting 75–105 year old buildings. Partner with NYC programs supporting historic property upgrades.	Property value appreciation of 15–30% post-renovation; taps largest building cohort
4	Mid-range pricing is most active (Insight #4)	Prioritize sourcing and listing inventory in the $300K–$600K price band. Develop tiered pricing models to attract first-time buyers and mid-range investors.	Maximizes transaction velocity; highest liquidity in this price segment
5	Seasonal peaks in Fall/Spring (Insight #5, #6)	Align marketing campaigns, open house schedules, and promotional budgets with Fall and Spring peaks. Reduce fixed operating costs during Winter months to improve margin.	10–15% improvement in conversion rates by synchronizing campaigns with buyer intent cycles

11. Limitations & Future Work

11.1 Limitations

- The dataset covers only 12 months (Sep 2016 – Aug 2017) and may not reflect longer-term trends or post-COVID market shifts
- Zero-price transactions (gifted or inter-family transfers) were excluded, which may undercount total market activity
- Missing square footage values required median imputation, introducing a small degree of estimation error
- The dataset does not include listing prices, days on market, or mortgage financing data, which are key drivers of buyer behavior
- Neighborhood-level granularity was available but not exhaustively analyzed due to scope constraints

11.2 Future Work

- Extend the dataset to a multi-year panel (2010–2024) to capture long-term price cycles and COVID-19 impact
- Incorporate crime statistics, school ratings, and transit proximity as explanatory variables in a regression model
- Build a predictive pricing model using gradient boosting (XGBoost / LightGBM) trained on property features
- Add census demographic data to identify gentrification patterns and displacement risk
- Automate the Tableau dashboard refresh pipeline using Tableau REST API

12. Tech Stack & Tools

Tool / Technology	Role	Status
Python 3.x + Jupyter Notebooks	ETL pipeline, data cleaning, statistical analysis, KPI computation	Mandatory
pandas, numpy	Data manipulation, aggregation, feature engineering	Mandatory
matplotlib, seaborn	Exploratory visualizations during EDA	Mandatory
scipy, statsmodels	Statistical testing and correlation analysis	Mandatory
Google Colab	Cloud-based notebook execution environment	Supported
Tableau Public	Interactive dashboard design, publishing, and sharing	Mandatory
GitHub	Version control, collaboration, and contribution audit trail	Mandatory
SQL (optional)	Initial data extraction and filtering	Optional

13. Team Contribution Matrix

The table below documents each team member's contribution across all project phases. Contributions are verifiable through GitHub Insights, pull request history, and committed artifacts.

Team Member	Data Sourcing	ETL/Cleaning	EDA	Statistical Analysis	Tableau	Report/PPT
Khushi	–	–	✓	✓	–	✓
Krishnadevan M.L.	✓	✓	–	–	–	–
Kanchan Rani	–	–	–	–	✓	–
Anuradha Raghuwanshi	–	–	–	✓	–	✓
Bhavya Goel	–	–	✓	–	–	–
Aditya Ranjan	–	✓	–	–	–	–

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Leads: Khushi & Krishnadevan M.L. | Date: 29 April 2026

14. Conclusion

This project successfully delivered a complete data analytics pipeline for New York City's real estate market, transforming 84,548 raw transaction records into 33,167 high-quality data points and surfacing actionable market intelligence through Python-based analysis and three interactive Tableau dashboards.

The analysis confirms that the NYC real estate market in 2016–2017 was characterized by strong geographic price differentiation (Manhattan vs. outer boroughs), overwhelming residential dominance (family homes at 88%+ of transactions), a mid-market pricing sweet spot ($300K–$600K), and seasonal volatility that peaks in Fall and Spring.

The five recommendations — luxury investment in Manhattan, residential marketing expansion, legacy building redevelopment, mid-range inventory prioritization, and campaign alignment with seasonal peaks — provide a data-backed strategic roadmap for investors, brokers, and analysts seeking to operate in this complex market.

The project team demonstrated competency across the full data analytics lifecycle: from problem framing and ETL engineering through statistical analysis, interactive visualization, and business storytelling — fulfilling all requirements of the Newton School of Technology Data Visualization & Analytics Capstone.